

Part III

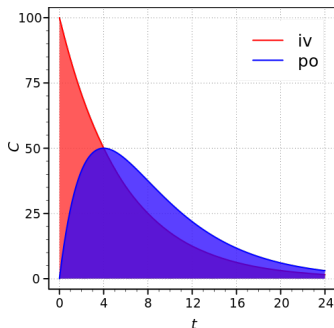
Integration techniques

Example: Bioavailability

Pharmacology: Bioavailability describes extent and rate with which an administered dose of unchanged drug reaches the systemic circulation.

Bioavailability is typically assessed by determining the “area under the plasma concentration–time curve” (AUC).

Absolute bioavailability compares the bioavailability of the active drug in systemic circulation following non-intravenous administration with the bioavailability of the same drug following intravenous administration.



Absolute bioavailability is the dose-corrected ratio of areas under the curves,

$$F_{abs} = 100 \frac{AUC_{po} \cdot D_{iv}}{AUC_{iv} \cdot D_{po}}$$

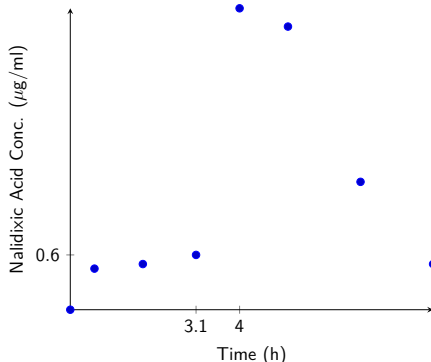
iv, intravenous; po, oral route. C is plasma concentration

Picture source: Wikipedia

Nalidixic Acid

Say we want to construct tablets of Nalidixic Acid (a synthetic antibiotic). However, doing this often results in a lower bioavailability of the active substance than administered intravenously.

To investigate this a 250 mg tablet of Nalidixic acid was administered to a patient, whereafter the concentration was measured.

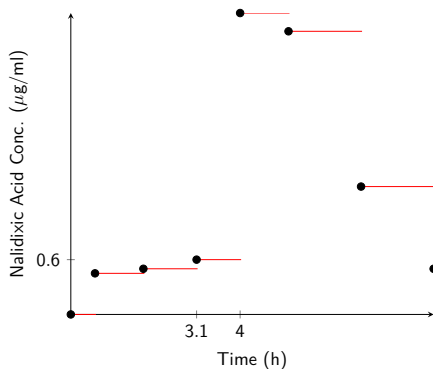


What is the area under the curve?

Area Under the Curve (AUC)

To answer this question we have to first interpolate the data and then calculate the AUC. We will present 3 ways:

- Constant interpolation, using left end points

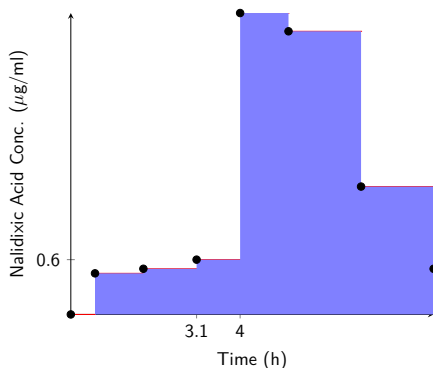


Area Under the Curve (AUC)

To answer this question we have to first interpolate the data and then calculate the AUC. We will present 3 ways:

- ▶ Constant interpolation, using left end points

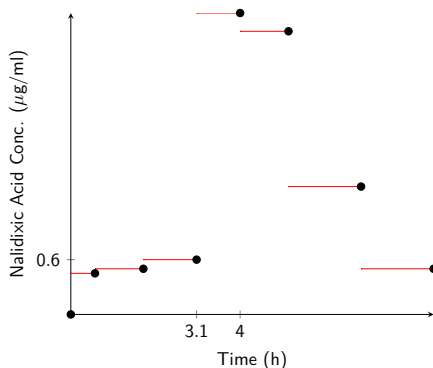
⇒ **Left Riemann Sum**
 $AUC = 11.6 \mu\text{g} \cdot \text{h}/\text{ml}$



Area Under the Curve (AUC)

To answer this question we have to first interpolate the data and then calculate the AUC. We will present 3 ways:

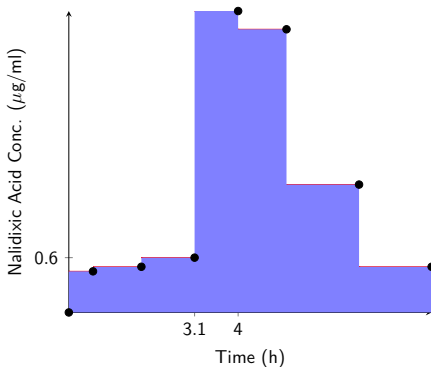
- ▶ Constant interpolation, using left end points
- ⇒ Left Riemann Sum
 $\text{AUC} = 11.6 \mu\text{g} \cdot \text{h}/\text{ml}$
- ▶ Constant interpolation, using right end points



Area Under the Curve (AUC)

To answer this question we have to first interpolate the data and then calculate the AUC. We will present 3 ways:

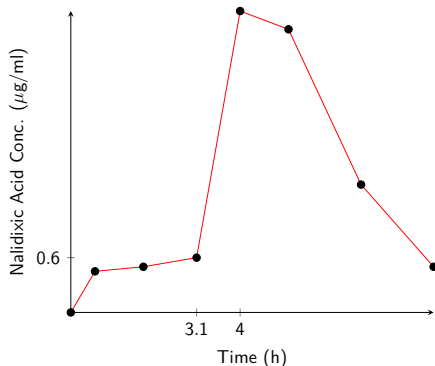
- ▶ Constant interpolation, using left end points
⇒ Left Riemann Sum
AUC = $11.6 \mu\text{g} \cdot \text{h}/\text{ml}$
- ▶ Constant interpolation, using right end points
⇒ Right Riemann Sum
AUC = $10.6 \mu\text{g} \cdot \text{h}/\text{ml}$



Area Under the Curve (AUC)

To answer this question we have to first interpolate the data and then calculate the AUC. We will present 3 ways:

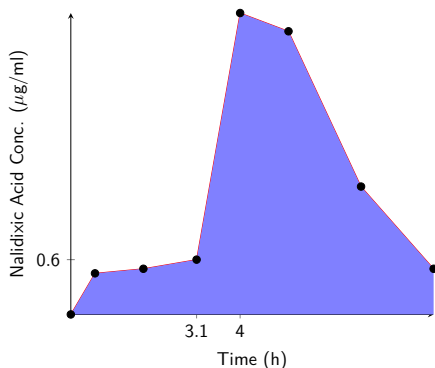
- ▶ Constant interpolation, using left end points
⇒ Left Riemann Sum
AUC = $11.6 \mu\text{g} \cdot \text{h}/\text{ml}$
- ▶ Constant interpolation, using right end points
⇒ Right Riemann Sum
AUC = $10.6 \mu\text{g} \cdot \text{h}/\text{ml}$
- ▶ **Linear Interpolation**



Area Under the Curve (AUC)

To answer this question we have to first interpolate the data and then calculate the AUC. We will present 3 ways:

- ▶ Constant interpolation, using left end points
⇒ Left Riemann Sum
AUC = $11.6 \mu\text{g} \cdot \text{h}/\text{ml}$
- ▶ Constant interpolation, using right end points
⇒ Right Riemann Sum
AUC = $10.6 \mu\text{g} \cdot \text{h}/\text{ml}$
- ▶ Linear Interpolation
⇒ Trapezoidal
AUC = $11.1 \mu\text{g} \cdot \text{h}/\text{ml}$

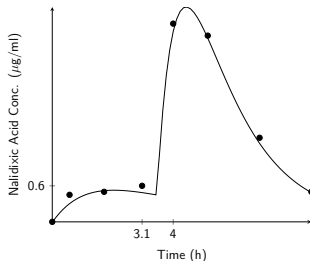


From Data to Function

After having performed the bioavailability experiment several times, the following model has been fitted to the observed data:

$$f(t) = \sum_{i=1}^n \alpha_i (e^{\beta_i(t-T_i)} - e^{\gamma_i(t-T_i)}) \quad \text{for } t \in [T_{n-1}, T_n] \quad (1)$$

For $n = 2$ we obtain the following fit, see ref. for parameters:



AUC for Functions

Thus we are now interested in the area under the curve given by f

This is defined to be the definite integral (Note: the outcome is a number):

$$\int_a^b f(x)dx \quad (2)$$

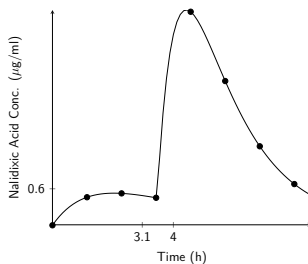
However, often it is not possible to calculate integrals explicitly

IDEA: Use the function to introduce new “datapoints” and use the techniques from before to approximate the integral!

From Function to Data

We proceed as follows:

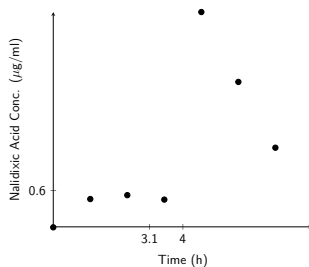
- Evaluate the function in n (usually equidistant) points,



From Function to Data

We proceed as follows:

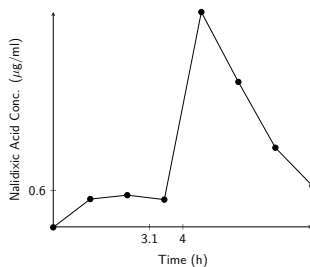
- ▶ Evaluate the function in n (usually equidistant) points,
- ▶ Ignore the old function,



From Function to Data

We proceed as follows:

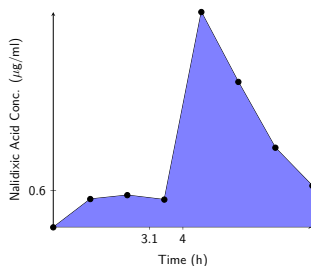
- ▶ Evaluate the function in n (usually equidistant) points,
- ▶ Ignore the old function,
- ▶ Interpolate the points (using either Left/Right Riemann or Trapezoidal),



From Function to Data

We proceed as follows:

- ▶ Evaluate the function in n (usually equidistant) points,
- ▶ Ignore the old function,
- ▶ Interpolate the points (using either Left/Right Riemann or Trapezoidal),
- ▶ Calculate the area under the resulting curve.



AUC for Functions

Thus we are now interested in the area under the curve given by f

This is defined to be the definite integral (Note: the outcome is a number):

$$\int_a^b f(x)dx \quad (2)$$

NOW: Exact calculation of the integral

Recap of Introduction to Integration

- ▶ Last time we introduced the definite and the indefinite integral of a function f .

Recap of Introduction to Integration

- ▶ Last time we introduced the definite and the indefinite integral of a function f .
- ▶ The definite integral was of the form $\int_a^b f(x)dx$ and gives, for positive f , the area under the curve f .

Recap of Introduction to Integration

- ▶ Last time we introduced the definite and the indefinite integral of a function f .
- ▶ The definite integral was of the form $\int_a^b f(x)dx$ and gives, for positive f , the area under the curve f .
- ▶ The indefinite integral $\int f(x)dx$ is the family of all antiderivatives of f .

Recap of Introduction to Integration

- ▶ Last time we introduced the definite and the indefinite integral of a function f .
- ▶ The definite integral was of the form $\int_a^b f(x)dx$ and gives, for positive f , the area under the curve f .
- ▶ The indefinite integral $\int f(x)dx$ is the family of all antiderivatives of f .
- ▶ Remember that F is an antiderivative of f if $F'(x) = f(x)$.

Recap of Introduction to Integration

- ▶ Last time we introduced the definite and the indefinite integral of a function f .
- ▶ The definite integral was of the form $\int_a^b f(x)dx$ and gives, for positive f , the area under the curve f .
- ▶ The indefinite integral $\int f(x)dx$ is the family of all antiderivatives of f .
- ▶ Remember that F is an antiderivative of f if $F'(x) = f(x)$.
- ▶ Integration and differentiation are inverse operations, i. e.

$$\frac{d}{dx} \int_0^x f(t)dt = f(x).$$

- ▶ Today we will develop tools to integrate a huge class of functions.

Substitution method

Example

Suppose the concentration of a drug in the bloodstream, $C(t)$, is given by

$$C(t) = e^{\beta(t-T)}$$

Determine the total exposure to the body, i. e., the AUC (area under the curve) of the drug concentration profile.

Substitution method

Let F be an antiderivative to f .
chain rule:

$$\frac{d}{dx}F(g(x)) = F'(g(x))g'(x) = f(g(x))g'(x).$$

Substitution method

Let F be an antiderivative to f .
chain rule:

$$\frac{d}{dx}F(g(x)) = F'(g(x))g'(x) = f(g(x))g'(x).$$

$$\implies \int f(g(x))g'(x)dx = F(g(x)) + c.$$

Substitution method

Let F be an antiderivative to f .

chain rule:

$$\frac{d}{dx}F(g(x)) = F'(g(x))g'(x) = f(g(x))g'(x).$$

$$\implies \int f(g(x))g'(x)dx = F(g(x)) + c.$$

Example

Consider

$$\int \sin(x)^2 \cos(x)dx$$

and choose $f(x) := x^2$ and $g(x) := \sin(x)$.

Substitution method

$$\int f(g(x))g'(x)dx = F(g(x)) + c$$

With the auxiliary variable $u = g(x)$, it follows that $\frac{du}{dx} = g'(x)$, and we can write

$$\int f(u)du = F(g(x)) + c$$

Substitution method

$$\int f(g(x))g'(x)dx = F(g(x)) + c$$

With the auxiliary variable $u = g(x)$, it follows that $\frac{du}{dx} = g'(x)$, and we can write

$$\int f(u)du = F(g(x)) + c$$

Example

- Consider

$$\int \frac{x}{1+x^2} dx.$$

- Choose $u := x^2 + 1$. Then $\frac{du}{dx} = 2x$.

Substitution method

$$\int f(g(x))g'(x)dx = F(g(x)) + c$$

With the auxiliary variable $u = g(x)$, it follows that $\frac{du}{dx} = g'(x)$, and we can write

$$\int f(u)du = F(g(x)) + c$$

Example

- Consider

$$\int \frac{x}{1+x^2} dx.$$

- Choose $u := x^2 + 1$. Then $\frac{du}{dx} = 2x$.
- Formal manipulation:

$$\frac{du}{dx} = 2x \implies du = 2x dx \implies \frac{1}{2} du = x dx$$

Substitution method

$$\int f(g(x))g'(x)dx = F(g(x)) + c$$

With the auxiliary variable $u = g(x)$, it follows that $\frac{du}{dx} = g'(x)$, and we can write

$$\int f(u)du = F(g(x)) + c$$

Example

- Consider

$$\int \frac{x}{1+x^2} dx.$$

- Choose $u := x^2 + 1$. Then $\frac{du}{dx} = 2x$.
- Formal manipulation:

$$\frac{du}{dx} = 2x \implies du = 2x dx \implies \frac{1}{2} du = x dx$$

- Now replace $u = 1 + x^2$ and $\frac{1}{2} du = x dx$ in the integral:

$$\int \frac{x}{1+x^2} dx = \int \frac{1}{u} \frac{1}{2} du = \frac{1}{2} \ln |u| + C = \frac{1}{2} \ln |1+x^2| + C = \frac{1}{2} \ln(1+x^2) + C.$$

Substitution method

Example

Suppose the concentration of a drug in the bloodstream, $C(t)$, is given by

$$C(t) = e^{\beta(t-T)}$$

Determine the total exposure to the body, i. e., the AUC (area under the curve) of the drug concentration profile.

Substitution method

Example

Suppose the concentration of a drug in the bloodstream, $C(t)$, is given by

$$C(t) = e^{\beta(t-T)}$$

Determine the total exposure to the body, i. e., the AUC (area under the curve) of the drug concentration profile.

The substitution method is often effective if one can spot a function and an expression similar to its derivative.

Substitution method

Example

Suppose the concentration of a drug in the bloodstream, $C(t)$, is given by

$$C(t) = e^{\beta(t-T)}$$

Determine the total exposure to the body, i. e., the AUC (area under the curve) of the drug concentration profile.

The substitution method is often effective if one can spot a function and an expression similar to its derivative.

Example (Poll everywhere)

What is $\int 3xe^{x^2} dx$?

Integration by parts

Just as the substitution method is a counterpart to the chain rule, partial integration is a counterpart to the product rule:

$$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$$

Integration by parts

Just as the substitution method is a counterpart to the chain rule, partial integration is a counterpart to the product rule:

$$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$$

$$\Rightarrow \int f(x)g'(x)dx = f(x)g(x) - \int g(x)f'(x)dx$$

Integration by parts

Just as the substitution method is a counterpart to the chain rule, partial integration is a counterpart to the product rule:

$$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$$

$$\Rightarrow \int f(x)g'(x)dx = f(x)g(x) - \int g(x)f'(x)dx$$

Example

Consider $\int x \cos(x)dx$.

Integration by parts

Just as the substitution method is a counterpart to the chain rule, partial integration is a counterpart to the product rule:

$$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$$

$$\Rightarrow \int f(x)g'(x)dx = f(x)g(x) - \int g(x)f'(x)dx$$

Example

Consider $\int x \cos(x)dx$.

Example (Poll everywhere)

What is

$$\int_0^1 3xe^x dx?$$

Example (Ex. 23.12, Substitution and Integration by Parts)

Find $\int e^{\sqrt{x}} dx$.

Example (Ex. 23.12, Substitution and Integration by Parts)

Find $\int e^{\sqrt{x}} dx$.

Example (Ex. 23.13, Using Integration by Parts Twice)

Find $\int e^x \sin(x) dx$.